

ranges covered by Qi, Abhishek Kumar, business development manager, Power Management Products, Texas Instruments (India), explains, "The focus for the lower power range is up to 5W and for the medium power range is up to 15W. Some examples of low-power products are applications using a single-cell battery, mobile phones, digital cameras, toys, Bluetooth headsets, audio headsets, alarm clocks, etc. Examples of mid-power products include tablets, ultra books, high-end smartphones, two-way radios, toys, industrial and medical equipment like barcode scanners, POS systems, health monitors, etc. Apart from this, Qi is also working on kitchen appliances and is defining specifications for devices like blenders, cooktops, coffee makers and others, that can be powered wirelessly." TI offers multi-mode wireless charging chips that support Qi as well as PMA specs.

Powermat's technology (which is the basis of the PMA's specifications) is also quite similar to Qi. It, too, works using tightly-coupled inductive charging, but the main difference is that it works on the 277-357kHz band, while WPC's Qi works on the 100-205kHz band. The PMA provides a comprehensive suite of standards and protocols covering inductive power technology, resonant power technology (in progress), digital transceiver communication, cloud-based power management services and environmental sustainability. The specifications are available on the consortium's website.

One stark difference between Qi and Powermat is that, while one is backed mainly by manufacturers, the other is backed mainly by users. This is evident by the fact that the most influential board members of PMA include P&G, AT&T Mobility, Starbucks and Powermat. Powerkiss, an earlier Qi ally, has also now joined forces with the PMA, bringing with it large clients like McDonalds and The Coffee Bean and Tea Leaf. Delta Airlines' lobbies, the revamped Barclays' Centre, New York City's Madison Square Garden, and Starbucks outlets are some of the places where one can experience PMA charging technol-

Key technologies

- Wireless Power Consortium's Qi
- Power Matters Alliance's Power 2.0
- Alliance for Wireless Power A4WP specifications (based on Qualcomm's WiPower)
- Witricity (which will be incorporated into Power 2.0)

ogy first-hand. General Motors has also chosen PMA's technology for in-car wireless charging for its future models. DuPont Building Innovation has partnered with the PMA to embed wireless charging solutions undemure everyday surfaces like kitchen countertops, tables and desks. In short, the PMA is trying to become ubiquitous by wooing the users rather than the sellers of technology.

Another key reason why PMA is seen as giving Qi a run for its money is its focus on 'intelligence' and improving the user experience with cloud-based services. In February this year, for example, Duracell Powermat demonstrated wireless power surfaces that operate as a mesh network, allowing wireless charging spots across the globe to be centrally-monitored and managed. This new network topography is ideal for use in public places such as coffee shops, airports and arenas, enabling those who manage venues to monitor, upgrade and set policies for each wireless charging spot from a centralised cloud-based system. This transforms wireless power in public places from disjointed, standalone surfaces into a smart global network, allowing the implementation of new services and business models. For example, there could be apps that find the nearest available wireless charging spot; service providers could control who can use their wireless power, for how long and on what terms; enterprises can monitor usage patterns, assess the health of their wireless power network and administer support and upgrades remotely, etc.

A4WP promises spatial freedom

One of the other notable contenders in this standards race is the A4WP, a consortium backed by Qualcomm and Samsung, amongst others. The focus of this group

is to advance spatial freedom in wireless power. And so, the specifications they put out will take a different approach to wireless charging. The A4WP specification uses magnetic resonance to charge a device in close proximity (up to around 3.8 cm or 1.5 inches away). Thus, a tablet or smartphone can be charged when placed next to a laptop that sports resonance charging capabilities, without having to be placed directly on a charging platform.

According to an explanation by the president of A4WP in a media interview, "The specification is based on non-radiative, near-field magnetic resonance, or loosely-coupled wireless power transfer, defined in a CEA-published glossary of terms as resonant wireless transfer of power through magnetic induction between coil(s) where the coupling factor (k) can be less than 0.1, though values up to 1 may also be supported, and where the system requires (coupled) magnetic resonance. What makes the technology unique is that its application demands that the designers explicitly address impedance matching between the charger and device to be charged. In contrast, in the case of classic magnetic induction, or so-called tightly-coupled solutions, there is no need for impedance matching as the charger and device to be charged are held in extremely close proximity, where even the slightest rotational or translational shift causes power transfer and efficiency to drop significantly. Ultimately, this difference between tightly-coupled (a first-generation wireless power transfer technology) and loosely-coupled (a second-generation technology) is the improvement in user experience owing to spatial freedom."

Consumers will like A4WP because it supports simultaneous charging of multiple devices with widely differing power requirements such as handsets, Bluetooth headsets, MP3 players, GPS devices and mobile tablets. It also delivers spatial freedom, which means that users can drop multiple devices randomly onto the charging surface without bothering to precisely position them. It will also fit well into next-generation home healthcare and outpatient monitoring devices where, for reasons